博 士 学 位 论 文

动态记忆应力的生物力学研究及临床意义探讨

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Three-Dimensional Finite Element Analysis of Nitinol Patellar Concentrator and Its Clinical Significance

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Keywords: Nitinol Patellar Concentrator, Patellar Fracture, Three-Dimensional Finite Element Analysis

Abstract. The Nitinol Patellar Concentrator (NT-PC) was designed to treat patellar fractures. It has been successfully applied in clinic proved by a survey of 4023 cases till 1995. Three-dimensional finite element analysis for the NT-PC fixation has been carried out to show the features and advantages of the device. The results are reported in this paper.

Introduction

Patellar fractures are very common in accident, and there are several methods to treat them, but for the partial and total comminuted fractures, the results were not satisfactory. In 1986, in order to get better therapy results of patellar fractures, especially the comminuted fractures, we designed a new device made of NiTi shape memory alloy, named as Nitinol patellar concentrator (NT-PC), which exerts initiative and continuous compression for the fixation. This article reported the study of the three-dimensional (3-D) finite element analysis and clinical applications of the NT-PC.

Structure of NTC

The shape of the NT-PC was designed to consist of two basis patellae claws, three apex patellae claws and a conjunctive waist as shown in Fig.1 [1,2]. Nitinol plates containing 50.3~51.8at% Ni and 1.8~2.2mm in thickness were used to prepare different sizes of the NT-PC, which were treated to exhibit one way shape memory effect with recovery temperature of 30±2℃ and reversible deformation of 8%. In clinical operation, the claws is to be unfolded in aqua astricta, put on to the fractured patellar and warmed up, as shown in Fig.2. Under body temperature the NT-PC tends to recover to its original shape resulting in a recovery compressive force. The difference of radians between NT-PC and the patellar making the fixation of the NT-PC tightly on the comminuted patellar till the fracture heals up.

Fig.1 Structure of NT-PC: ① basis patellae claws, ② conjunctive waist and ③ apex patellae claws

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Biomechanical study of posterior wall acetabular fracture fixation using acetabular tridimensional memory alloy-fixation system

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ABSTRACT

Background: We developed the acetabular tridimensional memory alloy-fixation system (ATMFS), which is made of NiTi shape memory alloy, according to the specific mechanical properties of biological memory material, nitinol alloy and measured distribution of contact area and pressure between the acetabulum and the femoral head of cadaveric pelvis.

Methods: Six formalin-preserved cadaveric pelvis were used for this investigation. Pressure-sensitive film was used to measure contact area and pressure within the anterior, superior, and posterior regions of the acetabulum. The pelvis were loaded under the following four conditions: (1) intact; (2) following a creation posterior wall fracture defect; (3) following reduction and standard internal fixation with reconstruction plate; (4) following reduction and internal fixation with a new shape memory alloy device named ATMFS. A posterior wall fracture was created along an arc of 40–90° about the acetabular rim.

Findings: Creation of a posterior wall defect resulted in increased load in the superior acetabulum (1422 N) as compared to the intact condition (762 N, P = 0.007). Following reduction and internal fixation, the load distribu ... bulum (1480 N) was not statistically different from the defect condition. Follow ... TMFS, the load seen at the superior region of the acetabulum (936 N) was less ... struction plate and was not different from intact state (P = 0.4).

Interpretation: These data indicate that the use of ATMFS as a fracture internal fixation device resulted a partial restoration of joint loading parameters toward the intact state. ATMFS fixation may result in a clinical benefit.

1. Introduction

Fractures of the posterior wall are the most common of the acetabular fractures (Ebraheim et al., 2007; Moed and McMichael, 2008). Operative treatment of these fractures has produced varying results by different authors (Giannoudis et al., 2007; Petsatodis et al., 2007). One of the major goals in managing acetabular fractures is the prevention of posttraumatic arthritis. Unreduced fractures involving the weight bearing portion of the acetabulum lead to arthritis (Templeman et al., 1999). When fractures occur in the acetabulum, loading is significantly altered throughout the hip (Konrath et al., 1998b; Olson et al., 1997). These alterations in loading were observed even after a simulated posterior wall fracture is reduced and fixed anatomically with standard buttress plating techniques as previously described (Goulet et al., 1994; Olson et al., 1995). Posttraumatic hip arthritis has been reported in association with as many as 20% of posterior wall acetabular fractures treated with open reduction and internal fixation (Goulet et al., 1994; Matta, 1996). It has been postulated that increased joint contact stresses from incongruity or altered load distributions within the hip joint eventually leads to degenerative posttraumatic arthritis through repetitive cartilage damage (Hadley et al., 1990; Olson et al., 1995). Thus, improving articular fragment stabilization beyond that attainable with standard internal fixation techniques may result in decreased fragment motion and therefore improved load distribution.

NiTi shape memory alloy, as a functional metal material, has many advantages such as shape memory effect, remarkable resistance to wear and corrosion, as well as good histocompatibility. Since its approval into the medical market by the US Food and Drug Administration (FDA) in 1990, NiTi shape memory alloy has been applied extensively. NiTi shape memory alloy is regarded as an important discovery in medical material science and a valuable "biological memory material". The NiTi hemofilter designed by Simon was demonstrated to be effective following application in 147 patients at 19 cardiovascular centers (Levi et al., 2008). Later on, with further studies, shape memory alloys were gradually applied in clinical practices such as the department of stomatology or orthopedics, and gained huge success (Levi et al., 2008; Oh et al., 2006; Templeman et al., 1999).

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Biomechanical Comparison of Different Stabilization Constructs for Unstable Posterior Wall Fractures of Acetabulum. A Cadaveric Study

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Abstract

Purpose: Operative treatment of unstable posterior wall fractures of acetabulum has been widely recommended. This laboratory study was undertaken to evaluate static fixation strength of three common fixation constructs: interfragmentary screws alone, in combination with conventional reconstruction plate, or locking reconstruction plate.

Methods: Six formalin-preserved cadaveric pelvises were used for this investigation. A posterior wall fracture was created along an arc of 40–90 degree about the acetabular rim. Three groups of different fixation constructs (two interfragmentary screws alone; two interfragmentary screws and a conventional reconstruction plate; two interfragmentary screws and a locking reconstruction) were compared. Pelvises were axial loaded with six cycles of 1500 N. Dislocation of superior and inferior fracture site was analysed with a multidirectional ultrasonic measuring system. Results: No statistically significant difference was found at each of the superior and inferior fracture sites between the three types of fixation. In each group, the vector dislocation at superior fracture site was significantly larger than inferior one.

Conclusions: All those three described fixation constructs can provide sufficient stability for posterior acetabular fractures and allow early mobilization under experimental conditions. Higher posterior acetabular fracture line, transecting the weight-bearing surface, may indicate a substantial increase in instability, and need more stable pattern of fixation.

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Introduction

Fracture of the posterior wall is the most common acetabular fracture [1,2]. They accounted for nearly 47% of the total acetabular fractures in the study by Letournel and Judet [2]. Operative treatment of these fractures with an unstable hip or when a large part of the posterior wall is involved has been widely recommended for anatomical reduction and rigid fixation [1–4]. However, Operative treatment of these fractures has produced varying results by different authors. It has generally been perceived that isolated fractures of the posterior wall have a good outcome [2,4], but recent reviews have shown that 21% to 32% of patients have poor results [5–7]. Furthermore, only 82% of patients had a good to excellent result despite perfect reduction in 94% [2], and there is up to a 30% failure rate within one year after fixation of posterior wall fractures, even when surgically treated by experienced orthopaedic traumatologists [1]. As results, the gap or step in articular surface would induce the development of osteoarthritis and degeneration [2–5]. Moreover, redislocation is another severe complication of failure fixation [3–6,9].

Although the failures are multifactorial, instable fixation method with followed premature mobilization is associated with poorer outcome [8,10,11]. It is clear that early postoperative rehabilitation training is beneficial to joint function recovery [2–8]. Therefore, beside anatomic reduction, a rigid fracture fixation implant, which allows early mobilization and prevents secondary displacement with a need for subsequent hip arthroplasty, seems to be of further importance in treatment of those fractures. A variety of fixation methods have been described for posterior acetabular wall fracture [1,5–7,9–11]. Generally, surgical fixation of posterior wall acetabular fractures is accomplished with two or three interfragmentary screws alone [9,12] or in combination with a buttress plate [8,13]. Several investigators have attempted to examine the contact area and load distribution of intact, fractured, and repaired cadaveric posterior acetabular wall with different fixation methods [3,8,13,14]. Others evaluate the stability of different fixation types for anterior wall, transverse, both-column, or T-type acetabular fractures [15–20]. But it is surprising that few biomechanical studies have been done to identify optimum technique of fixation for posterior wall fracture. Whether it is rigid enough by using screws fixation alone for posterior wall fracture? Is it necessary to combine with a buttress plate? How much can the additional plate contribute to the fixation stability? Can the locking plate provide more stability than conventional

PATENTS：

序号	专利名称	授权号	发明人	授权时间	专利类别
1	一种取弹片系统及其方法	201518009794.3	许硕贵；何滨	2019年03月15日	国防发明专利
2	C-ARM X-RAY APPARATUS WITH SURG	10463326B2	Bin He;Shigui Yan;Shuogui Xu;Liping Shen	2019年11月05日	国际专利
3	FRACTURE DEDUCTION SYSTEM	11083496B2	Bin He; Shuogui Xu; Rui Tong; Yu Wu	2021年08月10日	国际专利
4	靶点路径的确定方法、装置和系统、可读存储介质	201910161263.3	沈丽萍 ;许硕贵 ;何滨 ;童睿 ;孙伟伟	2020年12月08日	发明专利
5	带有手术定位和线性导航功能的C形臂X光机	201510196237.6	何滨,严世贵,许硕贵,沈丽萍.	2016年10月05日	发明专利
6	基于生物打印技术的细胞-生物支架复合物的制备方法	201310250953.9	谢杨,孙晓飞,章云童,韩倞,毛宁方,杨宗德,倪海键,许硕贵,夏琰.	2014年07月09日	发明专利
7	基于无创式实时手术定位导航设备的骨科手术机器人	201610047353.6	何滨,沈丽萍,许硕贵,童睿.	2017年09月08日	发明专利
8	具有电响应特性的形状记忆材料及其制备方法	200410073393.5	朱光明 ;.秦瑞丰 ;.许硕贵 ;	2006年08月09日	发明专利
9	无创式实时手术定位3D导航设备	201610659818.3	许硕贵,何滨,沈丽萍	2019年01月04日	发明专利
10	无创式实时手术定位导航设备	201310515922.1	何滨,许硕贵.	2015年09月09日	发明专利
11	一种动物脊髓损伤造模装置	201410554225.1	谢杨,李超,孙晓飞,佘光辉,章云童,许硕贵,杨国士,张馨雯,夏琰.	2016年01月06日	发明专利
12	一种横向梯度孔结构的纳米人工骨支架及其制备方法	201110287964.5	许硕贵,周斌,夏琰,唐骏,杜艾,卢春闻,张猛.	2014年12月10日	发明专利
13	一种类似天然骨结构的纳米人工骨支架及其制备方法	201310432628.4	许硕贵,王红英,夏琰,张艳艳,周潘宇,程晓松.	2017年04月12日	发明专利
14	一种生物可吸收捆绑带及其形状记忆锁扣	200610023175.X	许硕贵;.朱光明 ;.张春才.;	2007年12月19日	发明专利
15	一种用于骨折治疗的智能个性化外骨骼装置	201610914541.4	许硕贵; 何滨 ;	2017年09月08日	发明专利
16	尺桡骨形状记忆接骨器	00216128.1	张春才,许硕贵,苟三怀,迟永辉,张德胜.	2001年02月28日	实用新型专利
17	大鼠腹部注射固定装置	201720230380.7	袁野 ;.李佳浓 ;.周潘宇 ;.许硕贵;.姜里强 ;.张洪跃 ;.吴江红 ;	2018年05月22日	实用新型专利
18	带有手术定位和线性导航功能的C形臂X光机	201520250205.5	何滨,严世贵,许硕贵,沈丽萍.	2015年09月09日	实用新型专利
19	防冷凝的呼吸软管及呼吸湿化设备	201920424104.3	姜立国 ;.许硕贵;	2020年03月17日	实用新型专利
20	肱骨远端镍钛记忆合金接骨器	02217427.3	张春才 ;张德胜 ;苟三怀 ;张雪松 ;.李重茂 ;.曲乐丰 ;.王芝芳 ;.许硕贵;	2003年05月07日	实用新型专利
21	股骨颈骨折空心螺钉导向装置	201820569864.9	许硕贵; 吴江红 ;.何滨 ;周潘宇 ;袁野 ;陈汉清 ;	2019年04月12日	实用新型专利
22	骨骼复位系统	201721567092.7	何滨 ;.许硕贵;.童睿 ;.吴雨 ;	2019年06月04日	实用新型专利
23	骨骼复位装置	201721568294.3	何滨 ;.许硕贵;.方华磊 ;.王婧 ;	2019年02月05日	实用新型专利
24	骨折复位及维持装置	201821279863.7	吴江红 ;.吴雨斐 ;.许硕贵;.周潘宇 ;夏伊玲 ;吴琦 ;夏德昕 ;张洪跃 ;	2019年10月11日	实用新型专利
25	基于C臂X光机的激光定位仪及手术导航系统	201920068139.8	许硕贵;.吴江红 ;.何滨 ;周潘宇 ;.夏德萌 ;.李磊 ;.汪洋 ;	2020年01月17日	实用新型专利
26	基于无创式实时手术定位导航设备的骨科手术机器人	201620068764.9	何滨,沈丽萍,许硕贵,童睿.	2016年08月17日	实用新型专利
27	肩锁关节弹性镍钛记忆合金固定器	02217424.9	张德胜 ;.许硕贵;	2003年05月07日	实用新型专利
28	具有升降功能的旋转式激光定位导航标定装置	201720852596.7	袁野 ;周潘宇 ;许硕贵;.吴江红 ;姜里强 ;张洪跃 ;白思蓉 ;徐金山 ;	2019年02月15日	实用新型专利
29	聚髌器撑开器	02215441.8	张德胜 ;张春才 ;许硕贵 ;梁洁 ;孙兰萍 ;.王芝芳 ;.苟三怀 ;	2002年12月11日	实用新型专利
30	可生物降解的血管吻合环/轮及其专用手术器械	201720605628.3	许硕贵; 汤焘 ;.肖亮 ;.严新谊 ;.王攀峰 ;.周潘宇 ;	2018年11月06日	实用新型专利
31	可吸收粗隆截骨固定组合器	201020525734.9	许硕贵,王吉广,方永刚,张春才.	2011年05月04日	实用新型专利
32	可吸收弓齿状记忆接骨器	201020525762.0	许硕贵,刘德鼎,徐斌,王吉广,张春才.	2011年06月29日	实用新型专利
33	可吸收髋臼后柱解剖记忆固定器	201020525783.2	许硕贵,徐斌,刘德鼎,张春才.	2011年09月14日	实用新型专利
34	可吸收髋臼前柱解剖记忆聚合器	201020525765.4	许硕贵,方永刚,卢春闻,张春才,朱光明.	2011年06月01日	实用新型专利
35	可吸收三维解剖型髋臼记忆内固定系统	201020525732.X	许硕贵,吴亚乐,华夏,张春才.	2011年06月01日	实用新型专利
36	实验用老鼠固定装置	201720230959.3	袁野 ;.周潘宇 ;.许硕贵;.李佳浓 ;.姜里强 ;.张洪跃 ;.吴江红 ;	2018年05月22日	实用新型专利
37	无创式实时手术定位3D导航设备	201620871979.4	许硕贵,何滨,沈丽萍	2017年06月27日	实用新型专利
38	下胫腓复位内固定镍钛合金记忆钩	02217422.2	张德胜 ;张雪松 ;许硕贵 ;孙兰平 ;	2003年07月02日	实用新型专利
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40	一种尺骨鹰嘴形状记忆内固定器	200820057266.X	王家林,许硕贵,张春才.	2009年02月04日	实用新型专利
41	一种带磁性的氧气筒安全警示牌	201620655794.X	胡敏,张玲,许硕贵,王群.	2017年03月08日	实用新型专利
42	一种带有缝合标记线的抗菌手术贴膜	201821279876.4	吴江红 ;.吴琦 ;.许硕贵;.周潘宇 ;.吴雨斐 ;夏伊玲 ;章云童 ;郝兹辰 ;	2019年10月08日	实用新型专利

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43	一种弹力绷带	201821152772.7	周潘宇 ;. 夏德萌 ;. 许硕贵;. 袁野 ;. 王胥人 ;. 熊文韬 ;. 孙大鹏 ;	2019年07月05日	实用新型专利
44	一种骨针折弯组件	201720389678.2	周潘宇 ;. 吴江红 ;. 许硕贵;. 姜里强 ;. 袁野 ;. 张洪跃 ;. 夏琰 ;. 王欢 ;	2018年10月12日	实用新型专利
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46	一种既可降温又可供饮水的双功能水冷降温马甲	201921807453.X	李磊 ;. 常文军 ;. 张馨丹 ;. 许硕贵;. 张凡 ;. 袁浩 ;	2020年08月14日	实用新型专利
47	一种具有气囊结构的注射针	201822008755.2	周潘宇 ; 夏德萌 ; 许硕贵 ; 倪喆鑫 ;. 李磊;. 王胥人 ;. 汪洋 ;. 王硕 ;	2019年09月24日	实用新型专利
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49	一种全地形担架	201420603360.6	谢杨, 章云童, 许硕贵, 孙晓飞, 马兵, 李超, 董微, 金光辉, 夏琰, 张世杰	2015年01月28日	实用新型专利
50	一种桡、尺动脉压迫止血器	201720418915.3	吴江红 ;. 周潘宇 ;. 许硕贵;. 朱嘉琦 ;. 袁野 ;. 姜里强 ;. 张洪跃 ;. 夏琰 ;	2018年10月12日	实用新型专利
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53	一种适用于指、掌骨骨折固定的钢板内固定装置	201220262946.1	谢杨, 李明, 陈自强, 许硕贵, 张春才, 孙晓飞.	2012年12月05日	实用新型专利
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55	一种腰椎固定带	201620677951.7	王立芬, 张玲娟, 席惠君, 张玲, 许硕贵,	2017年04月12日	实用新型专利
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57	一种医用开口器	201520867084.9	徐立, 胡敏, 张玲, 许硕贵, 张笑平, 祁智, 张敏伶	2016年05月04日	实用新型专利
58	一种用于骨折修复的骨骼复位机构	201621141453.7	许硕贵;. 何滨 ;	2017年09月26日	实用新型专利
59	一种用于骨折治疗的智能个性化外骨骼装置	201621141323.3	许硕贵, 何滨	2017年10月03日	实用新型专利
60	一种止血注射装置	201822009961.5	夏德萌 ;. 周潘宇 ;. 许硕贵;. 孔凯文 ;. 李磊 ;. 王胥人 ;. 张洪跃 ;. 汪洋 ;	2019年09月24日	实用新型专利
61	一种止血装置	201822008698.8	周潘宇 ;. 夏德萌 ;. 汪洋 ;. 许硕贵;. 章云童 ;. 李磊 ;. 王胥人 ;. 张洪跃 ;	2019年09月24日	实用新型专利
62	一种注射装置	201822011833.4	夏德萌 ;. 周潘宇 ;. 许硕贵;. 张洪跃 ;. 夏琰 ;. 李磊 ;. 王胥人 ;. 汪洋 ;	2019年12月17日	实用新型专利

63	用于防护用具的密封装置和防护设备	202020162585.8	杨峰 ;许硕贵 ;盛春泉 ;王丹 ;陆斌 ;邹晟 ;徐丽霞 ;王冶 ;沈素萍	2021年04月27日	实用新型专利
64	警示牌（带磁性的氧气筒安全警示牌）	201630284702.7	胡敏, 张玲, 许硕贵, 王群.	2016年12月14日	外观设计
65	密封贴	202030600245.4	杨峰 ;许硕贵 ;盛春泉 ;王丹 ;陆斌 ;邹晟 ;徐丽霞 ;王冶 ;沈素萍	2020年12月15日	外观设计
66	柄下骨折记忆加压接骨器	01344219.8	张春才, 徐卫东, 许硕贵.	2002年05月08日	外观设计
67	弓齿形状记忆接骨器（III型）	01344220.1	张春才, 许硕贵, 苏佳灿.	2002年05月08日	外观设计
68	弓齿形状记忆接骨器（II型）	01344221.X	张春才, 许硕贵, 苏佳灿.	2002年05月08日	外观设计
69	弓齿形状记忆接骨器（I型）	01344222.8	张春才, 许硕贵, 苏佳灿.	2002年05月08日	外观设计
70	解剖型记忆加压接骨器	01344218.X	张春才, 许硕贵, 孙满法.	2002年06月26日	外观设计
71	髋臼前柱解剖记忆固定器（II型）	01344223.6	张春才, 许硕贵, 苏佳灿, 张雪松.	2002年05月08日	外观设计
72	髋臼前柱解剖记忆固定器（I型）	01344224.4	张春才, 许硕贵, 苏佳灿, 张雪松.	2002年05月08日	外观设计
73	尺骨鹰嘴导向记忆固定器	01344217.1	张春才, 王家林, 许硕贵.	2003年02月05日	外观设计
74	尺桡骨形状记忆接骨器（Ⅱ型）	01352651.0	张春才, 许硕贵, 王家林, 吴玉堂.	2002年07月24日	外观设计
75	尺桡骨形状记忆接骨器（I型）	01352648.0	张春才, 许硕贵, 苏佳灿, 孙满法.	2003年01月01日	外观设计
76	后臼壁记忆固定器	01352649.9	张春才, 许硕贵, 苏佳灿, 孙满法.	2003年01月22日	外观设计
77	后柱臼解剖型记忆固定器	01352647.2	张春才 ; 许硕贵 ; 苏佳灿 ; 吴玉堂 ;	2002年06月26日	外观设计
78	记忆接骨器(髋臼后柱Ⅲ型)	200730146002.2	张春才 ;. 许硕贵 ;. 达国祖 ;	2008年03月19日	外观设计
79	接骨器(弓齿形状记忆型)	200730145895.9	张春才 ;. 许硕贵 ;. 达国祖 ;	2008年04月23日	外观设计
80	接骨器(天鹅式记忆型)	200730146005.6	张春才 ;. 许硕贵 ;. 达国祖 ;	2008年04月23日	外观设计
81	臼后壁记忆接骨器(Ⅰ型)	200730145896.3	张春才 ;. 许硕贵 ;. 达国祖 ;	2008年06月25日	外观设计
82	髋臼后柱记忆固定器	01352650.2	张春才, 许硕贵, 孙满法.	2002年10月16日	外观设计
83	髋臼后柱记忆接骨器(Ⅰ型)	200730145900.6	张春才 ;. 许硕贵 ;. 达国祖 ;	2009年03月25日	外观设计
84	髋臼前柱记忆接骨器(Ⅰ型)	200730145898.2	张春才 ;. 许硕贵 ;. 达国祖 ;	2008年03月19日	外观设计
85	髋臼前柱记忆接骨器(Ⅱ型)	200730145899.7	张春才 ;. 许硕贵 ;. 达国祖 ;	2008年03月19日	外观设计
86	形状记忆接骨器	01352415.1	张春才, 王家林, 许硕贵, 孙满法, 吴玉堂.	2002年10月16日	外观设计

REGISTRATION CERTIFICATE FOR MEDICAL DEVICE:


中华人民共和国
PEOPLE'S REPUBLIC OF CHINA
医疗器械注册证
REGISTRATION CERTIFICATE FOR MEDICAL DEVICE

注册号：国食药监械(准)字2010第3460861号

兰州西脉记忆合金股份有限公司：

你单位生产的大转子截骨记忆固定器，经审查，符合医疗器械产品市场准入规定，准许注册，自批准之日起有效期四年。

特此证明。

国家食品药品监督管理局
State Food and Drug Administration

附件：医疗器械产品生产制造认可表

No.1004345

Memory Fixator for Greater Trochanteric Osteotomy, Reg. No.: 3460861.


中华人民共和国
PEOPLE'S REPUBLIC OF CHINA
医疗器械注册证
REGISTRATION CERTIFICATE FOR MEDICAL DEVICE

注册号：国食药监械(准)字2010第3460862号

兰州西脉记忆合金股份有限公司：

你单位生产的臼后壁导向记忆固定器，经审查，符合医疗器械产品市场准入规定，准许注册，自批准之日起有效期四年。

特此证明。

国家食品药品监督管理局
State Food and Drug Administration

附件：医疗器械产品生产制造认可表

No.1004346

Memory Fixator for Posterior Acetabular Wall Guidance, Reg. No.: 3460862.


中华人民共和国
PEOPLE'S REPUBLIC OF CHINA
医疗器械注册证
REGISTRATION CERTIFICATE FOR MEDICAL DEVICE

注册号：国食药监械(准)字2010第3460482号

兰州西脉记忆合金股份有限公司：

你单位生产的记忆合金髌骨爪，经审查，符合医疗器械产品市场准入规定，准许注册，自批准之日起有效期四年。

特此证明。

国家食品药品监督管理局
State Food and Drug Administration

附件：医疗器械产品生产制造认可表

No.1002318

Memory Alloy Patellar Claw
Reg. No.: 3460482.


中华人民共和国
PEOPLE'S REPUBLIC OF CHINA
医疗器械注册证
REGISTRATION CERTIFICATE FOR MEDICAL DEVICE

注册号：国食药监械(准)字2010第3460860号

兰州西脉记忆合金股份有限公司：

你单位生产的带齿记忆箍环状固定器，经审查，符合医疗器械产品市场准入规定，准许注册，自批准之日起有效期四年。

特此证明。

国家食品药品监督管理局
State Food and Drug Administration

附件：医疗器械产品生产制造认可表

No.1004490

Serrated Memory Cerclage Ring Fixator , Reg. No.: 3460860.

REGISTRATION CERTIFICATE FOR MEDICAL DEVICE:

中华人民共和国
PEOPLE'S REPUBLIC OF CHINA
医疗器械注册证
REGISTRATION CERTIFICATE FOR MEDICAL DEVICE

注册号：国食药监械(准)字2010第3460863号

兰州西脉记忆合金股份有限公司：

你单位生产的髋臼后柱解剖型记忆固定器，经审查，符合医疗器械产品市场准入规定，准许注册，自批准之日起有效期四年。

特此证明。

国家食品药品监督管理局
State Food and Drug Administration

附件：医疗器械产品生产制造认可表

No.1004347

Anatomical Memory Fixator for Posterior Acetabular Column, Reg. No.: 3460863.

中华人民共和国
PEOPLE'S REPUBLIC OF CHINA
医疗器械注册证
REGISTRATION CERTIFICATE FOR MEDICAL DEVICE

注册号：国食药监械(准)字2010第3460858号

兰州西脉记忆合金股份有限公司：

你单位生产的髋臼前柱解剖型记忆固定器，经审查，符合医疗器械产品市场准入规定，准许注册，自批准之日起有效期四年。

特此证明。

国家食品药品监督管理局
State Food and Drug Administration

附件：医疗器械产品生产制造认可表

No.1004488

Anatomical Memory Fixator for Anterior Acetabular Column, Reg. No.: 3460858.

中华人民共和国
PEOPLE'S REPUBLIC OF CHINA
医疗器械注册证
REGISTRATION CERTIFICATE FOR MEDICAL DEVICE

注册号：国食药监械(准)字2010第3460857号

兰州西脉记忆合金股份有限公司：

你单位生产的天鹅记忆固定器，经审查，符合医疗器械产品市场准入规定，准许注册，自批准之日起有效期四年。

特此证明。

国家食品药品监督管理局
State Food and Drug Administration

附件：医疗器械产品生产制造认可表

No.1004487

Swan Memory Fixator, Reg. No.: 3460857.

查新结论

上海长海医院委托之查新项目“髋臼骨折治疗的基础与临床研究”，经CBMDISC（1978-2010，5）和MEDLINE（1991-2010，5）光盘数据库检索及文献追踪：

髋臼位置深在，解剖关系复杂，髋臼骨折为高速高能量损伤的关节内骨折，易导致多种并发损伤的出现，延误诊治。因此髋臼骨折能否得到及时合理的诊治是确保其预后良好的关键。目前切开复位内固定已经成为移位性髋臼骨折的标准治疗方法，其治疗目标在于髋臼的解剖复位和坚强内固定，从而使早期活动成为可能。Judet-Letoumel的髋臼二柱概念和骨折分类及手术入路，已成为髋臼骨折治疗的里程碑【1-3】。髋臼骨折的内固定多采用钢板和螺钉，其中有采用钛钢板和镍钛记忆合金钉【4-5】。

关于髋臼骨折手术后的恢复效果，国外文献6报道，对60个平均随访5年的髋臼骨折外科治疗的患者评估疗效，其解剖复位率是62%，临床结果优良率为80%。文献7报道，髋臼骨折后路入路的外科治疗长期随访（10-15年）后，优良率为80%。

该课题组在分析研究髋臼的解剖与生物力学结构基础上，提出了髋臼的三柱理论，建立了髋臼骨折浮动分类方法，并以此为基础，设计了髋臼三维记忆内固定系统，展开了系列基础研究与临床研究。临床应用解剖复位率 92.7%，功能优良率 92.69%，明显高出国外水平，为解决复杂粉碎的髋臼骨折提供了一种有效的新方法。国内外未见与该课题组相同的文献报道。

查新单位 上海市医学科技查新委员会 （盖章） 上海市科学技术成果查新专用章（1） 2010 年 9 月 19 日

备注： 本结论仅供专家参考

Basic and Clinical Research on Acetabular Fracture Treatment. Anatomical reduction rate improved from 62% to 92.7%; functional excellent rate from 80% to 92.69%, surpassing international benchmarks. Conclusion: No identical prior literature found domestically or internationally. Issued by: Changhai Hospital, Shanghai, 2010.09.19.

许硕贵教授课题组发明的系列记忆接骨器临床应用例数统计（2014-2024）

年份	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	合计
例数	56500	71333	72578	80119	81226	94097	95683	65024	78149	77995	78452	851186
终端销售额（万元）	47963	62351	64489	72269	73564	86959	87447	57826	70746	70279	51345	745239
备注：按已推广应用到的 3225 家医院统计												

兰州西脉记忆合金股份有限公司

2025 年 2 月 6 日

Series of Shape Memory Bone Fixators. Applied in 3,225+ hospitals globally, including Johns Hopkins University and other international institutions. Issued by: Lanzhou Ximai Memory Alloy Co., Ltd., 2025.02.06.

Page 09

Clinical Reviews in Allergy & Immunology

Allergy to Surgical Implants

Karin A. Pacheco

Abstract

Surgical implants are essential elements of repair procedures to correct worn out joints, damaged spinal components, heart and vascular disease, and chronic pain. However, many of the materials that provide stability, flexibility, and durability to the implants are also immunogenic. Fortunately, allergic responses to surgical implants are infrequent. [illegible]

Keywords Surgical implant · Joint replacement · Joint failure · Rash · Metal allergy · Nickel · Adhesive allergy · Methyl methacrylate · Contact dermatitis

Introduction

Surgical implants rely on a number of materials selected on the basis of their strength, stability, pliability, durability, and other physical characteristics. Unfortunately, some of these materials also have allergenic properties that can cause immunological reactions and implant failure for a small number of patients. [illegible]

Published online: 15 September 2018

Annals of Biomedical Engineering

BMES

ORIGINAL ARTICLE

Determining the Mechanical Properties of Super-Elastic Nitinol Bone Staples Through an Integrated Experimental and Computational Calibration Approach

Dario Carbonaro · Claudio Chiastra · Federico A. Bologna · Alberto L. Audenino · Mara Terzini

Received: 27 July 2023 / Accepted: 1 December 2023
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Abstract

Super-elastic bone staples have emerged as a safe and effective alternative for internal fixation. Nevertheless, several biomechanical aspects of super-elastic staples are still unclear and require further exploration. [illegible]

Keywords Bone staple · Orthopedic device · Nickel–titanium · Experimental testing · Digital twin · Material identification

Introduction

Bone staples are internal fixation devices that are commonly used in the foot, hand, and ankle to provide bone alignment and stabilization after fracture or fusion procedures. [illegible]

Published online: 27 December 2023

Hindawi
International Journal of Biomaterials
Volume 2019, Article ID 4307461, 11 pages

Hindawi

Research Article

Combustion Synthesis Porous Nitinol for Biomedical Applications

H. Aihara,[1] J. Zider,[1] G. Fanton,[2] and T. Duerig[3]

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[2]Department of Orthopedic Surgery, Sports Medicine Division, Stanford University Medical Center, Palo Alto, California, USA
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Academic Editor: Alexander Seifalian

Porous Nitinol with a three-dimensional anisotropic interconnective open pore structure has been successfully produced by the combustion synthesis (CS) of elemental Ni and Ti powders. [illegible]

1. Introduction

Since the early 1960s, orthopedic implants were made from three distinct materials types, metals, ceramics, and polymers. [illegible]

Nitinol has widely been used for decades in cardiovascular, neurological and orthopedic devices. [illegible]

国家科学技术进步奖

证　书

为表彰国家科学技术进步奖获得者，特颁发此证书。

项目名称：动态记忆应力促进战创伤骨愈合的关键技术研究与临床转化

奖励等级：二等

获 奖 者：许硕贵

中华人民共和国国务院

2016年12月21日

证书号：2016-J-24400-2-02-R01

China National Science and Technology Progress Award

Title: Key Technology Research and Clinical Translation of Dynamic Memory Stress for Promoting War Trauma Bone Healing

报告编号：20202101C0701102

科 技 查 新 报 告

项目名称：手术激光定位导航系统

委 托 人：中国人民解放军海军军医大学第一附属医院
上海长海医院

委托日期：二〇二〇年四月二十八日

查新机构：中国科学院上海科技查新咨询中心
（科技查新专用章）

查新完成日期：二〇二〇年五月十一日

中 华 人 民 共 和 国 科 学 技 术 部

二〇一六年制

查新项目名称	中文：手术激光定位导航系统			
	英文：Surgical laser positioning and navigation system			
查新机构	名 称	中国科学院上海科技查新咨询中心		
	通信地址	上海市岳阳路 319 号 31 号楼 C 座底楼	邮编	200031
	负 责 人	江洪波	联系人	刘 佩
	电 话	021-54922920	传 真	021-54922934
	电子信箱	chaxin@sibs.ac.cn	网 址	http://www.chaxin.ac.cn

一、查新目的

成果

二、查新项目的科学技术要点

创新点：

项目组（上海长海医院和杭州三坛医疗科技有限公司）发明了基于 C 型臂 X 光机（C 臂机）的以空间智能可视为特征的手术激光定位导航新装备。项目组根据 C 臂机成像原理，首次构建了 C 臂机透视成像"数学圆锥几何模型"，首次提出了"体内目标成像射线空间可视"的全新手术导航模式，通过原创设计"激光管透镜微调机构、基准板叠加配打基准孔特种加工法"等硬件技术和"定位模型算法、图像识别、畸变校准等"等软件技术，并通过上千次实验，最终发明了手术激光定位导航系统，并实现了体内病灶的毫米级定位精度和手术路径智能化可视三维导航。

技术指标：

定位精度高，误差小于 1.5mm；定位迅速，单次定位时间小于 5s，有效减少了患者和手术医护人员的辐射暴露。

三、查新点与查新要求

查新点：

手术激光定位导航系统

1. 首次构建了 C 臂机透视成像"数学圆锥几何模型"（或称"同心圆定位原理"）。

2. 首次提出了"体内目标成像射线空间可视化模式"（英文称"X-ray Trajectory Visualization during Target Imaging"）的全新手术导航模式。

3. 原创设计"激光管透镜微调机构、基准板叠加配打基准孔特种加工法"等硬件技术和"定位模型算法、图像识别、畸变校准等"等软件技术，实现定位误差小于 1.5mm，单次定位时间小于 5s，从而提高了手术操作精度，减少了手术创伤，缩短了手术时间及降低了患者和手术医护人员的辐射暴露。

查新要求：

查国内外有无相同或类似文献与研究报道，并作对比分析及新颖性判断。

国外相关文献中，主要内容有：附件 9 介绍了基于 x 线误差校正的股骨颈空心螺钉植入微创定位机器人系统；附件 10 介绍了完全集成的针定位系统的 c 臂系统与 ct 透导微创干预。

根据上述检索结果对比分析，发现公开的技术主要有微创激光导航系统的临床应用前研究、颅内血肿微创双 X 定位法的研究、机器人辅助微创手术视觉导航定位系统研究、基于 C 型臂 X 线机的激光定位导航技术在外科手术中的应用、光笔助 X 射线体表定位在骨科植入物内固定中的应用、无创式实时手术定位导航设备、附属于 C 臂 X 光机的手术定位导航设备、用于 C 形臂 X 射线机的透视导航方法等。检索发现，公开技术在技术特征和实现的技术路径上均存在明显区别，因而未见与本项目技术特征完全一致的公开技术。

综上所述，项目方关于"手术激光定位导航系统"其创新之处主要体现在：1. 首次构建了 C 臂机透视成像"数学圆锥几何模型"（或称"同心圆定位原理"）；2. 首次提出了"体内目标成像射线空间可视化模式"（英文称"X-ray Trajectory Visualization during Target Imaging"）的全新手术导航模式；3. 原创设计"激光管透镜微调机构、基准板叠加配打基准孔特种加工法"等硬件技术和"定位模型算法、图像识别、畸变校准等"等软件技术。实现定位导航精度≤0.1mm,临床使用整体精度≤1.0mm；单次定位时间小于 5s；从而提高了手术操作精度，减少了手术创伤、缩短了手术时间及降低了患者和手术医护人员的辐射暴露。经检索，国内外公开文献中未见与该项目技术特点完全相同的技术公开。因此，该项目具有新颖性。

查新员（签字）：　　查新员职称：馆员
审核员（签字）：　　审核员职称：副研究员（国家级审核员）

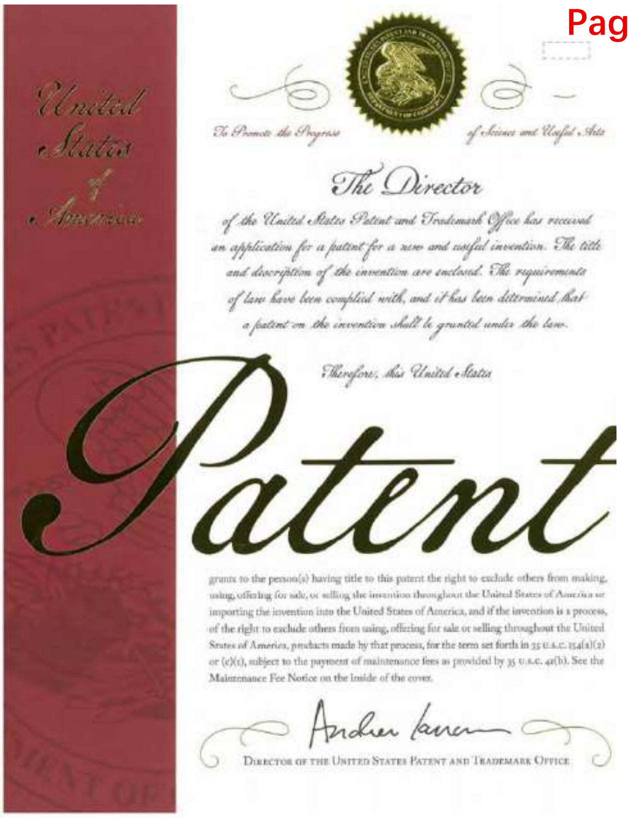
（科技查新专用章）

2020 年 5 月 11 日

Surgical Laser Positioning and Navigation System. Positioning accuracy ≤0.1mm (clinical overall accuracy ≤1.0mm); single positioning time <5s; effectively reduces radiation exposure for patients and surgical staff. Conclusion: No technically identical prior literature found domestically or internationally. Issued by: Shanghai Science and Technology Information Consulting Center, Chinese Academy of Sciences, 2020.05.11.

序号	发明名称	专利类别	专利号	所有发明人
1	无创式实时手术定位导航设备	发明专利	ZL201310515922.1	何滨、<u>许硕贵</u>
2	带有手术定位和线性导航功能的C形臂X光机	发明专利	ZL201510196237.6	何滨、严世贵、<u>许硕贵</u>、沈丽萍
3	基于无创式实时手术定位导航设备的骨科手术机器人	发明专利	ZL201610047353.6	何滨、沈丽萍、<u>许硕贵</u>、童睿
4	无创式实时手术定位3D导航设备	发明专利	ZL201610659818.3	<u>许硕贵</u>、何滨、沈丽萍
5	靶点路径的确定方法、装置和系统、可读存储介质	发明专利	ZL201910161263.3	沈丽萍、<u>许硕贵</u>、何滨、童睿、孙伟伟
6	一种取弹片系统及其方法	国防发明专利	ZL201518009794.3	<u>许硕贵</u>、何滨
7	C-ARM X-RAY APPARATUS WITH SURGICAL POSITIONING AND LINEAR NAVIGATION FUNCTION	美国发明专利	US10463326B2	何滨、严世贵、<u>许硕贵</u>、沈丽萍
8	带有手术定位和线性导航功能的C形臂X光机	实用新型专利	ZL201520250205.5	何滨、严世贵、<u>许硕贵</u>、沈丽萍.
9	基于无创式实时手术定位导航设备的骨科手术机器人	实用新型专利	ZL201620068764.9	何滨、沈丽萍、<u>许硕贵</u>、童睿
10	无创式实时手术定位3D导航设备	实用新型专利	ZL201620871979.4	<u>许硕贵</u>、何滨、沈丽萍
11	具有升降功能的旋转式激光定位导航标定装置	实用新型专利	ZL201720852596.7	袁野、周潘宇、<u>许硕贵</u>、吴江红、姜里强、张洪跃、白思嘉、徐金山
12	基于C臂X光机的激光定位仪及手术导航系统	实用新型专利	ZL201920068139.8	<u>许硕贵</u>、吴江红、何滨、周潘宇、夏德萌、李磊、汪洋

1. <u>*Shuogui Xu*</u>, Jianghong Wu, Bin He, Panyu Zhou, Demeng Xia, Lei Li and Yang Wang. Laser Positioning Deviceand Surgical Navigation System Based on C-arm X-ray Machine. Patent No.: ZL201920068139.8, 2020.01.17.
2. Liping Shen, <u>*Shuogui Xu**</u>, Bin He, Rui Tong and Weiwei Sun. Method, Device and System for Target Path Determination, and Readable Storage Medium. Patent No.: ZL201910161263.3, 2020.12.08.
3. Bin He, Shigui Yan, <u>*Shuogui Xu**</u> and Liping Shen. C-arm X-ray Apparatus With Surgical Positioning and Linear Navigation Function. Patent No.: US 10463326B2, 2019.11.05.
4. <u>*Shuogui Xu**</u>, Bin He. System and Method for Extracting Bullet Fragments. Patent No.: ZL201518009794.3, 2019.03.15.
5. Ye Yuan, Panyu Zhou, <u>*Shuogui Xu**</u>, Jianghong Wu, Liqiang Jiang, Hongyue Zhang, Sijia Bai and Jinshan Xu. Rotary Laser Positioning and Navigation Calibration Device with Lifting Function. Patent No.: ZL201720852596.7, 2019.02.15.
6. <u>*Shuogui Xu**</u>, Bin He and Liping Shen. Non-invasive Real-time Surgical Positioning 3D Navigation Device. Patent No.: ZL201610659818.3, 2019.01.04.
7. Bin He, Liping Shen, <u>*Shuogui Xu**</u> and Rui Tong. An Orthopedic Surgical Robot Based on Non-invasive Real-time Surgical Positioning and Navigation Device. Patent No.: ZL201610047353.6, 2017.09.08.
8. <u>*Shuogui Xu**</u>, Bin He and Liping Shen. Non-invasive Real-time Surgical Positioning 3D Navigation Device. Patent No.: ZL201620871979.4, 2017.06.27.
9. Bin He, Shigui Yan, <u>*Shuogui Xu**</u> and Liping Shen. C-arm X-ray Apparatus with Surgical Positioning and Linear Navigation Function. Patent No.: ZL201510196237.6, 2016.10.05.
10. Bin He, Liping Shen, <u>*Shuogui Xu**</u> and Rui Tong. An Orthopedic Surgical Robot Based on Non-invasive Real-time Surgical Positioning and Navigation Device. Patent No.: ZL201620068764.9, 2016.08.17.
11. Bin He, <u>*Shuogui Xu**</u>. Non-invasive Real-time Surgical Positioning and Navigation Device. Patent No.: ZL201310515922.1, 2015.09.09.
12. Bin He, Shigui Yan, <u>*Shuogui Xu**</u> and Liping Shen. C-arm X-ray Apparatus With Surgical Positioning and Linear Navigation Function. Patent No.: ZL201520250205.5, 2015.09.09.



US010463326B2

(12) **United States Patent**
He et al.

(10) **Patent No.: US 10,463,326 B2**
(45) **Date of Patent: Nov. 5, 2019**

(54) **C-ARM X-RAY APPARATUS WITH SURGICAL POSITIONING AND LINEAR NAVIGATION FUNCTION**

(71) Applicant: **HANGZHOU SANTAN MEDICAL TECHNOLOGY CO., LTD**, Zhejiang (CN)

(72) Inventors: **Bin He**, Zhejiang (CN); **Shigui Yan**, Zhejiang (CN); **Shuogui Xu**, Zhejiang (CN); **Liping Shen**, Zhejiang (CN)

(73) Assignee: **HANGZHOU SANTAN MEDICAL TECHNOLOGY CO., LTD**, Zhejiang (CN)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 195 days.

(21) Appl. No.: **15/560,162**

(22) PCT Filed: **May 29, 2015**

(86) PCT No.: **PCT/CN2015/080233**
§ 371 (c)(1),
(2) Date: **Sep. 21, 2017**

(87) PCT Pub. No.: **WO2016/169099**
PCT Pub. Date: **Oct. 27, 2016**

(65) **Prior Publication Data**
US 2018/0092614 A1 Apr. 5, 2018

(30) **Foreign Application Priority Data**
Apr. 23, 2015 (CN) 2015 1 0196237
Apr. 23, 2015 (CN) .... 2015 2 0250205 U

(51) **Int. Cl.**
A61B 6/12 (2006.01)
A61B 6/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***A61B 6/4441*** (2013.01); ***A61B 6/4405*** (2013.01); ***A61B 90/37*** (2016.02);
(Continued)

(58) **Field of Classification Search**
CPC A61B 6/4441; A61B 6/4405; A61B 6/00; A61B 6/12; A61B 90/37
See application file for complete search history.

(56) **References Cited**

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5,483,572 A 1/1996 Hoornaert et al.
2017/0296273 A9* 10/2017 Brown A61B 90/13

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CN 201642054 U 11/2010
(Continued)

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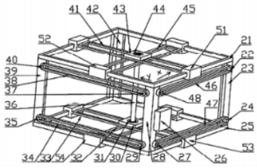
"International Search Report (Form PCT/ISA/210)", dated Aug. 26, 2015, with English translation thereof, pp. 1-4.

Primary Examiner — David P Porta
Assistant Examiner — Shun Lee
(74) *Attorney, Agent, or Firm* — JCIPRNET

(57) **ABSTRACT**

A C-arm X-ray apparatus with a surgical positioning and linear navigation function is provided. The apparatus includes a C-arm, an image enhancer, an X-ray tube, a support frame, a trolley, a monitor and a central controller. The image enhancer is disposed at one end of the C-arm, and the X-ray tube is disposed at the other end of the C-arm which is mounted on the support frame and capable of rotationally sliding along the support frame, the support frame is mounted on the trolley and capable of sliding horizontally along the trolley, and the monitor and the central controller are electrically connected with the image enhancer and the X-ray tube, where a positioning module is
(Continued)



中华人民共和国医疗器械注册证

注册证编号：国械注准20243010898

注册人名称	杭州三坛医疗科技有限公司
注册人住所	浙江省杭州市西湖区三墩镇西园九路8号3幢E座7楼701室
生产地址	浙江省杭州市西湖区三墩镇西园九路8号3幢E座7楼701室
代理人名称	/
代理人住所	/
产品名称	脊柱外科手术导航定位系统
型号、规格	MS-001
结构及组成	由规划模块、导引模块（含机械臂）、工具包、体位反馈模块组成。
适用范围	用于成人脊柱外科胸椎和腰椎手术中手术器械和植入物的导航定位。
附　件	产品技术要求
其他内容	/
备　注	/

审批部门：国家药品监督管理局

批准日期：二〇二四年五月十四日
生效日期：二〇二四年五月十四日
有效期至：二〇二九年五月十三日

中华人民共和国医疗器械注册证

注册证编号：国械注准20233210361

注册人名称	杭州三坛医疗科技有限公司
注册人住所	浙江省杭州市西湖区三墩镇西园九路8号3幢E座7楼701室
生产地址	浙江省杭州市西湖区三墩镇西园九路8号3幢E座7楼701室
代理人名称	/
代理人住所	/
产品名称	腰椎置钉手术计划软件
型号、规格	PV-001，发布版本1.0.0.27
结构及组成	由软件安装文件组成，功能模块包括：DICOM导入模块、规划模块、规划拓展模块、反馈模块、导出模块。
适用范围	用于成人腰椎椎弓根螺钉置钉手术的规划和评估。
附　件	产品技术要求
其他内容	/
备　注	/

审批部门：国家药品监督管理局

批准日期：二〇二三年三月二十一日
生效日期：二〇二三年三月二十一日
有效期至：二〇二八年三月二十日

中华人民共和国医疗器械注册证

注册证编号：浙械注准20172240913

注册人名称	杭州三坛医疗科技有限公司
注册人住所	杭州市西湖区三墩镇西园八路11号2幢4楼402室
生产地址	杭州市西湖区三墩镇西园八路11号2幢4楼402室
代理人名称	不适用
代理人住所	不适用
产品名称	激光定位系统
型号、规格	SY－001
结构及组成	产品由激光驱动器、主控PC、信号分流器和软件组成。
适用范围	激光定位系统需与C形臂X光机配合使用，用于骨骼肌肉系统手术中病灶点的定位，为手术入路提供参考。
附 件	产品技术要求
其他内容	/
备 注	

审批部门：浙江省食品药品监督管理局

批准日期：2017 年 08 月 15 日
有效期至：2022 年 08 月 14 日
（审批部门盖章）

中华人民共和国医疗器械注册证

注册证编号：浙械注准20192210492

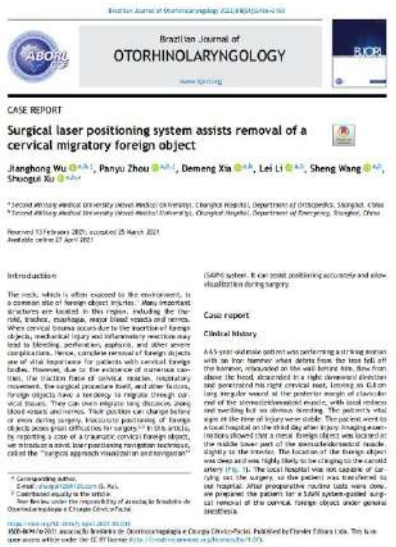
注册人名称	杭州三坛医疗科技有限公司
注册人住所	浙江省杭州市西湖区三墩镇西园九路8号3幢E座7楼701室
生产地址	浙江省杭州市西湖区三墩镇西园九路8号3幢E座7楼701室
代理人名称	不适用
代理人住所	不适用
产品名称	医学图像处理软件
型号、规格	YX-001
结构及组成	产品由软件和软件载体组成，软件载体为光盘；软件由DICOM模块，三维显示模块，划线功能模块组成。
适用范围	产品用于对来源于单模式或多模式的医学影像进行处理。
附　件	产品技术要求
其他内容	/
备　注	

审批部门：浙江省药品监督管理局

批准日期：[illegible] 23 日
有效期至：[illegible] 日

1. Surgical Navigation and Positioning System for Spinal Surgery (Class III Medical Device), Reg. No.: 20243010898.
2. Lumbar Screw Placement Surgical Planning Software (Class III Medical Device), Reg. No.: 20233210361.
3. Laser Positioning System (Class II Medical Device), Reg. No.: 20172240913.
4. Medical Image Processing Software (Class II Medical Device), Reg. No.: 20192210492.

Page 15


Brazilian Journal of
OTORHINOLARYNGOLOGY
CASE REPORT
Surgical laser positioning system assists removal of a cervical migratory foreign object
Jianghong Wu, Panyu Zhou, Demeng Xia, Lei Li, Sheng Wang, Shuogui Xu
Introduction
Case report
Clinical history


ORIGINAL ARTICLE
Novel laser positioning navigation to aid puncture during percutaneous nephrolithotomy: a preliminary report
Abstract
Introduction


Taylor & Francis
ORIGINAL ARTICLE
Assessment of radiation exposure in balloon kyphoplasty using a new navigation system
Bin He, Shujun Xie, Jing Wang, Chao Xu, Yingdelong Mao and Shuogui Xu
ABSTRACT
ARTICLE HISTORY
KEYWORDS
Introduction


TRICK OF THE TRADE
A novel technique for minimally invasive removal of a foreign body in the rectal wall
Introduction
Technique
Surgical procedure


Royal College of Surgeons
ONLINE CASE REPORT
Removing a metal foreign object successfully from a patient's retroperitoneal space using laparoscopy and a novel navigation system
KEYWORDS
Introduction
Case history

证书

国科奖社证字第0123号

为表彰发明创业奖创新奖获得者，特颁发此证书。

项目名称：手术激光定位导航新装备、新技术及临床应用

奖励等级：二等奖

完成单位：海军军医大学第一附属医院
杭州三坛医疗科技有限公司

完 成 人：许硕贵 吴江红 何 滨 周潘宇
章云童 汪 洋

证书号：2023-CAICX-2-JD09

中国发明协会
二〇二三年八月

China Association of Inventions Entrepreneurship Award Innovation Award

Title: Surgical Laser Positioning Navigation New Equipment, New Technology and Clinical Application

获奖证明

兹证明以下作品，参加中国国际大学生创新大赛（2025），并获得全国总决赛金奖。

作品名称：Integrated C-arm X-ray and Laser Technology for High-Precision Real-Time Surgical Positioning and Navigation

参赛学生：Shalong Xiong、Sebastian Gruber、Jinbo Ren、Hans Ewigmann、Yuanhao Wei、Minghao He、Rong Cha

指导教师：董雪、许硕贵、张暐

特此说明。

上海交通大学学生创新中心

2025 年 11 月 6 日

证书号第17624685号

实用新型专利证书

实用新型名称：一种耳塞式核心体温计

发 明 人：许硕贵;张丰

专 利 号：ZL 2022 2 1652946.2

专利申请日：2022年06月28日

专 利 权 人：中国人民解放军海军军医大学第一附属医院

地 址：200082 上海市杨浦区长海路168号、虹口区东江湾路15号

授权公告日：2022年10月21日　　授权公告号：CN 217637703 U

国家知识产权局依照中华人民共和国专利法经过初步审查，决定授予专利权，颁发实用新型专利证书并在专利登记簿上予以登记。专利权自授权公告之日起生效。专利权期限为十年，自申请日起算。

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国家知识产权局

2022年10月21日

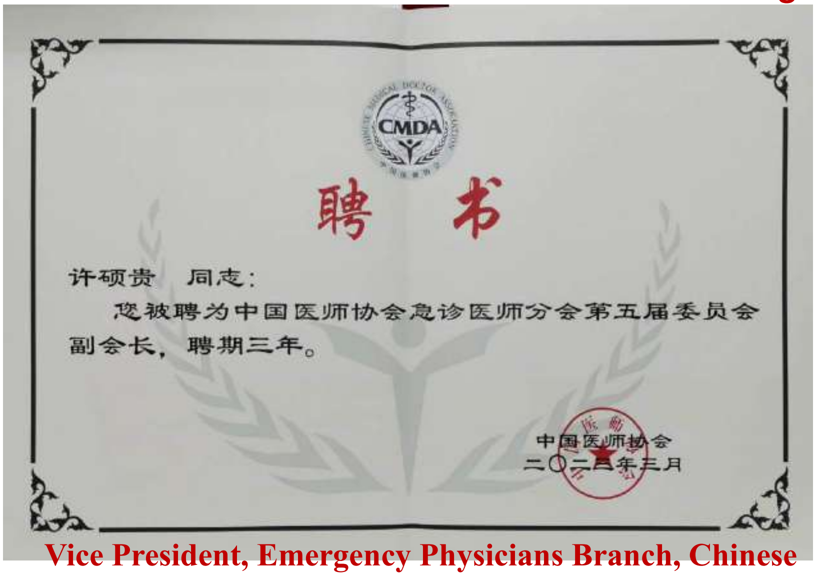
第1页（共2页）

其他事项参见续页

1. <u>*Shuogui Xu**</u>, Feng Zhang. A Wearable In-Ear Core Body Thermometer. Patent No.: ZL 202221652946.2, 2022.10.21.

Shanghai Emerging Science and Technology Collaborative Innovation Competition Award

Title: Ear Canal Implantable Core Body Temperature Monitor



聘 书

许硕贵　同志：

您被聘为中国医师协会急诊医师分会第五届委员会副会长，聘期三年。

中国医师协会
二〇二三年三月

Vice President, Emergency Physicians Branch, Chinese Medical Doctor Association

证书

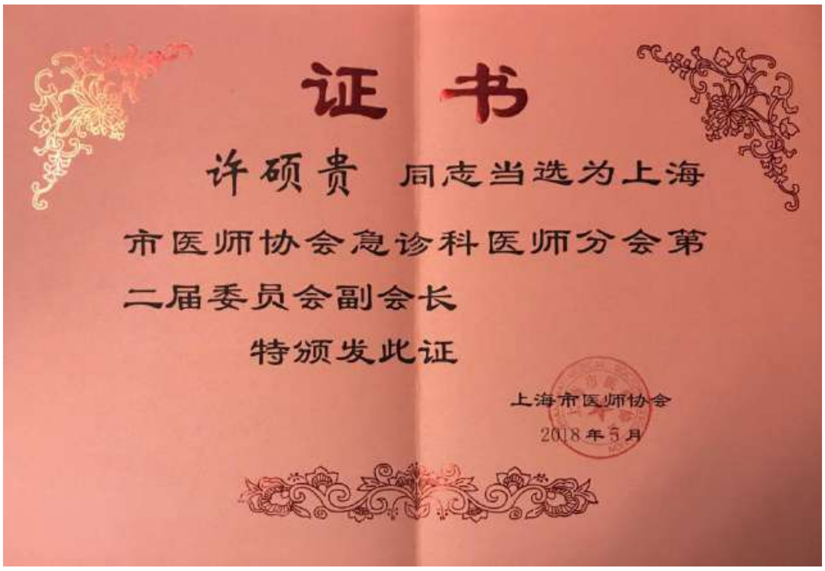
许硕贵　同志：

被推选为上海市医学会急诊医学专科分会第九届委员会副主任委员。

特颁发此证。

上海市医学会
二〇一九年十二月

Vice Chair, Emergency Medicine Subspecialty Branch, Shanghai Medical Association.



证　书

许硕贵　同志当选为上海市医师协会急诊科医师分会第二届委员会副会长

特颁发此证

上海市医师协会
2018年5月

Vice President, Emergency Physicians Branch, Shanghai Medical Doctor Association.

荣誉证书

许硕贵 同志：

经研究，聘请您为《山东医药》北京编辑委员会委员，任期2010~2012年。

二〇一〇年三月

Shandong Medical Journal

聘　书

兹聘请　许硕贵　教授为《中华创伤骨科杂志》第三届编辑委员会通讯编委。

中华医学会杂志社
二〇一一年四月十九日

中華醫學會

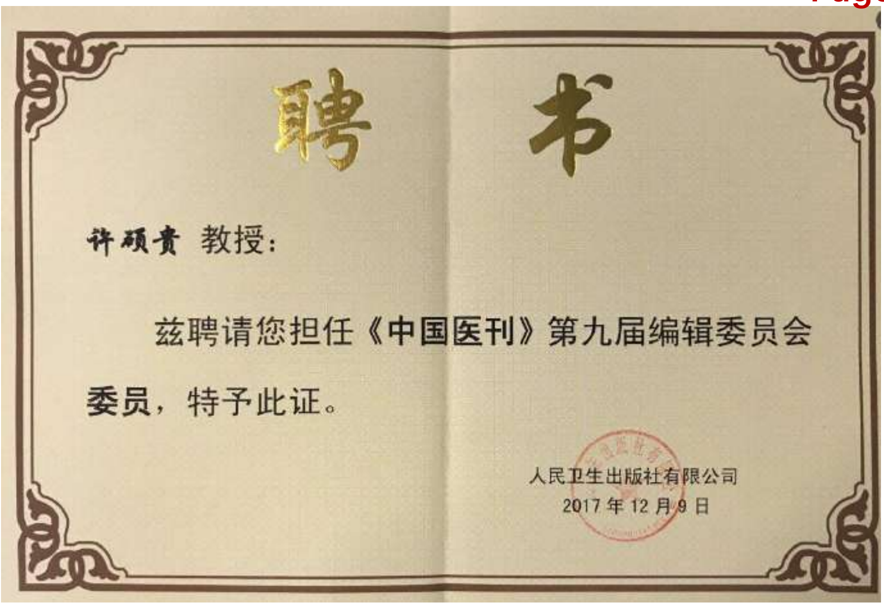
Chinese Journal of Orthopaedic Trauma

聘　书

兹聘请 许硕贵 同志为《中国急救医学》杂志第八届编辑委员会 编委 ，聘期叁年（从2016年6月至2019年5月）。

中国医师协会
黑龙江省科学技术情报研究院
2016年6月

Chinese Journal of Critical Care Medicine



聘书

许硕贵 教授：

兹聘请您担任《中国医刊》第九届编辑委员会委员，特予此证。

人民卫生出版社有限公司
2017年12月9日

Chinese Journal of Medicine

聘书

兹聘请 许硕贵 教授为《中华急诊医学杂志》第五届编辑委员会编辑委员。

中华医学会
二〇一九年十一月十九日

中華醫學會

Chinese Journal of Emergency Medicine

聘书

LETTER OF APPOINTMENT

Dear Professor Shuo-gui Xu,

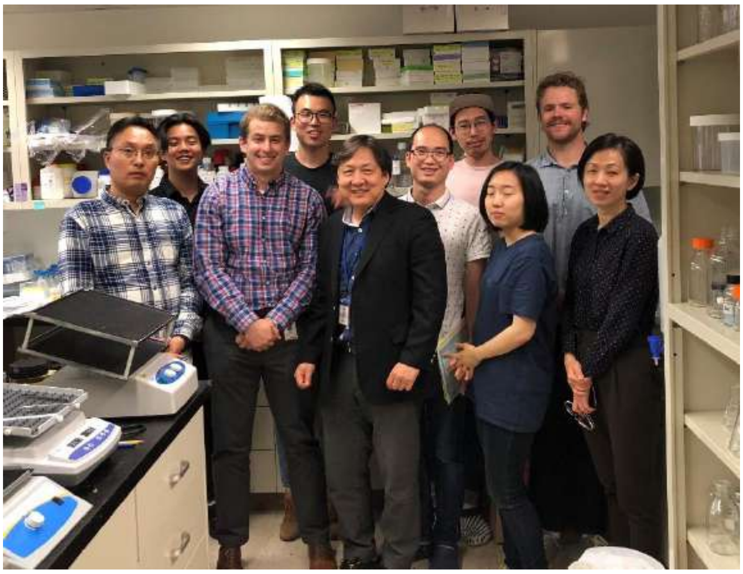
World Journal of Emergency Medicine (ISSN 1920-8642) is an international journal published in English by Zhejiang University. The scope of WJEM encompasses all clinical activities concerned with acute medical care. It has been indexed by major indexing database including: Science Citation Index Expanded (SCIE), PubMed, Pubmed Central, Google Scholar, Chemical Abstracts, VINITI database and EMBASE and EMCare, EBSCO, Index Copernicus, J-Gate, CNKI.

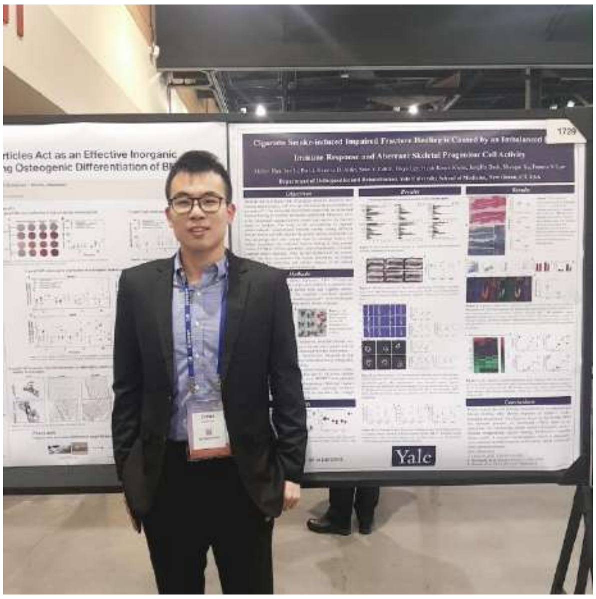
In recognition of your outstanding reputation and contribution in your field, it is our honor to invite you to be a member of editorial committee, WJEM. Your appointment will be from July 2019 to July 2021. We look forward to working with you and thank you very much in advance for your important contributions to the journal.

Sincerely yours,

Editorial Committee, *World Journal of Emergency Medicine*

Photos taken during the applicant's visits to the University of Louisville, Wake Forest University, and Stanford University School of Medicine in the United States





Photos taken during the applicant's doctoral student's joint training period at Yale University, and a photo of the two together

Clinical Reviews in Allergy & Immunology

Allergy to Surgical Implants

Karin A. Pacheco[1,2]

Abstract

Surgical implants are essential elements of repair procedures to correct worn out joints, damaged spinal components, heart and vascular disease, and chronic pain. However, many of the materials that provide stability, flexibility, and durability to the implants are also immunogenic. Fortunately, allergic responses to surgical implants are infrequent. When they do occur, however, the associated pain, swelling, inflammation, and decreased range of motion can significantly impair the implant function. Given the high numbers of joint replacements performed in the developed world, allergic reactions to orthopedic implants form the largest category of allergic responses. The most important allergens in this category include nickel, cobalt, chromium, and bone cement. These allergens are also the most important in reactions to spinal surgeries. Multiple cardiac and neurostimulatory devices are constructed of metals and adhesives that can be sensitizing in some individuals. Implantable pulse generators, important in cardiac pacemakers, gastric stimulators, and neurostimulators, may include components made of stainless steel, titanium alloy, platinum and iridium, epoxy resins, poly methyl methacrylates, and isocyanates, all of which are immunogenic in some patients. Cardiac stents and patches are often made of Nitinol, a composite of nickel and titanium. More surgical procedures are closed using skin glues, which are also capable of triggering a blistering contact dermatitis.

Introduction

Annals of Biomedical Engineering

BMES BIOMEDICAL ENGINEERING SOCIETY

ORIGINAL ARTICLE

Determining the Mechanical Properties of Super-Elastic Nitinol Bone Staples Through an Integrated Experimental and Computational Calibration Approach

Dario Carbonaro[1] · Claudio Chiastra[1] · Federico A. Bologna[1] · Alberto L. Audenino[1] · Mara Terzini[1]

Abstract

Super-elastic bone staples have emerged as a safe and effective alternative for internal fixation. Nevertheless, several biomechanical aspects of super-elastic staples are still unclear and require further exploration. Within this context, this study presents a combined experimental and computational approach to investigate the mechanical characteristics of super-elastic staples. Two commercially available staples with distinct geometry, characterized by two and four legs, respectively, were examined. Experimental four-point bending tests were conducted to evaluate staple performance in terms of generated forces. Subsequently, a finite element-based calibration procedure was developed to capture the unique super-elastic behavior of the staple materials. Finally, a virtual bench testing framework was implemented to separate the effect of geometry from that of the material characteristics on the mechanical properties of the devices, including generated force, strain distribution, and fatigue behavior. The experimental tests indicated differences in the force vs. displacement curves between staples.

Introduction

Hindawi

Research Article

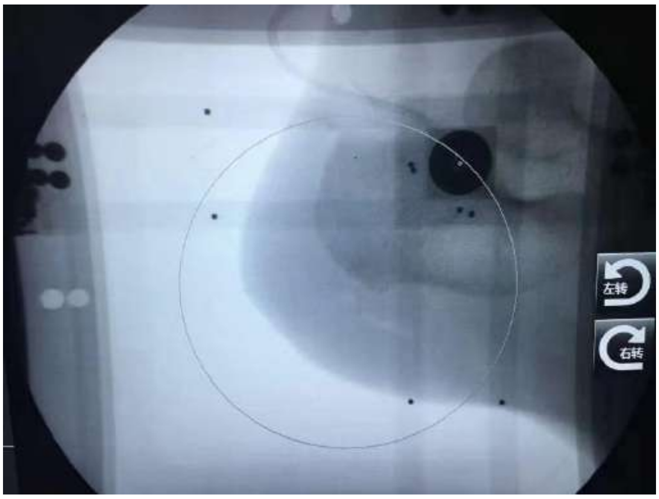
Combustion Synthesis Porous Nitinol for Biomedical Applications

H. Aihara,[1] J. Zider,[1] G. Fanton,[2] and T. Duerig[3]

Academic Editor: Alexander Setiahan

Porous Nitinol with a three-dimensional anisotropic interconnective open pore structure has been successfully produced by the combustion synthesis (CS) of elemental Ni and Ti powders. The resulting product can be tailored to closely match the stiffness of cancellous bone to minimize stress shielding.

1. Introduction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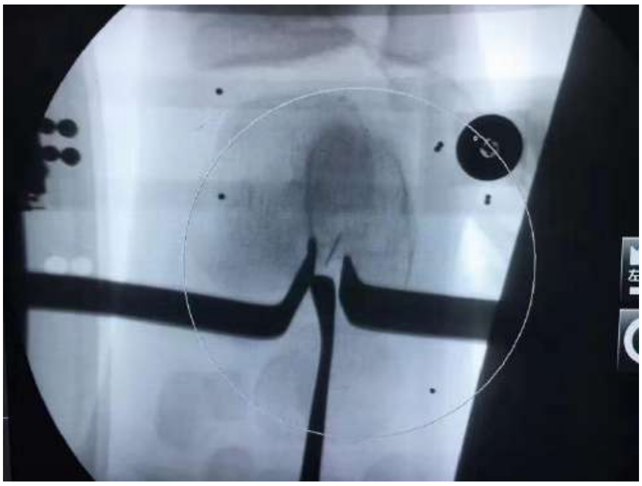

World Journal of Urology
https://doi.org/10.1007/s00345-018-2496-z

ORIGINAL ARTICLE

Novel laser positioning navigation to aid puncture during percutaneous nephrolithotomy: a preliminary report

Jianghong Wu[1] · Panyu Zhou[1,2] · Xi Luo[3] · Zichen Hao[1] · Chaoyue Lu[4] · Hongyue Zhang[1] · Tie Zhou[4] · Shuogui Xu[1,2]

Received: 25 July 2018 / Accepted: 17 September 2018

Abstract

Purpose Accurate puncture of the renal collecting system is crucial to the success of percutaneous nephrolithotomy and presents a technical challenge for urologists. Here, we introduced the Surgical Approach Visualization and Navigation (SAVN) system, a novel navigation system to assist puncture and reduce intraoperative radiation.

Materials and methods Twenty kidneys of 10 cadavers were randomly divided into two groups for renal calyx puncture. In the control group, traditional fluoroscopy was used for guidance, while SAVN system was used in the experimental group. Puncture duration, number of puncture attempts, total number of intraoperative fluoroscopies, and number of fluoroscopies during the puncture procedure were recorded.

Results The puncture duration was 14.2 ± 2.5 s in SAVN group and 48.3 ± 7.1 s in conventional group ($P < 0.05$). One puncture attempt was needed for successful puncture in SAVN group, while more than one in conventional group ($P = 0.28$). The total number of intraoperative fluoroscopies was 3.3 ± 1.0 in SAVN group and 14.5 ± 3.1 in control group ($P < 0.05$), while the number of fluoroscopies during the puncture procedure was 0 and 11.2 ± 2.4, respectively ($P < 0.05$).

Conclusions The novel SAVN system has a simplified structure and is easy to use. It can be used to successfully assist with puncture of the renal calyx, thus reducing puncture duration and radiation dose.

Keywords Kidney stones · Percutaneous nephrolithotomy · Puncture · Navigation

Introduction

Since its introduction in 1976, percutaneous nephrolithotomy (PCNL) has become the gold standard for the treatment of kidney stones larger than 20 mm in diameter [1], as it is minimally invasive and associated with few complications.

Jianghong Wu, Panyu Zhou and Xi Luo contributed equally to the article.

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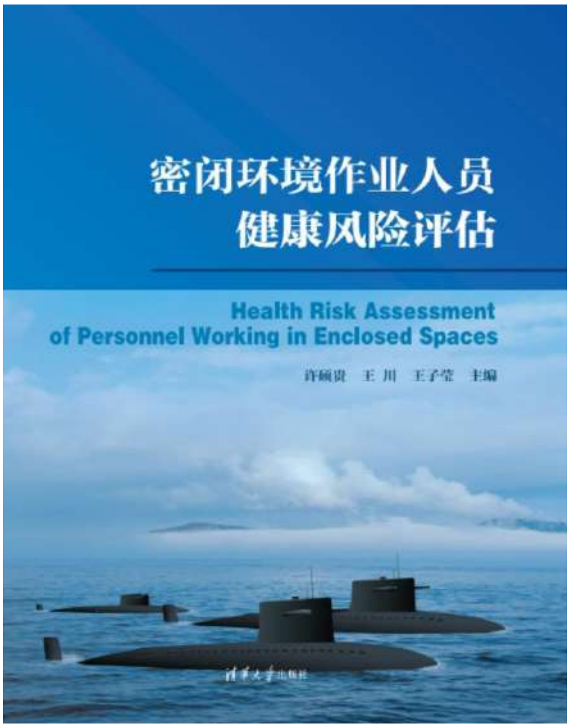
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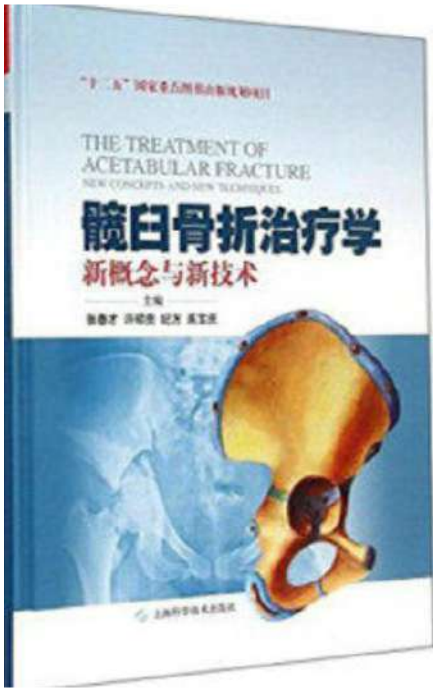
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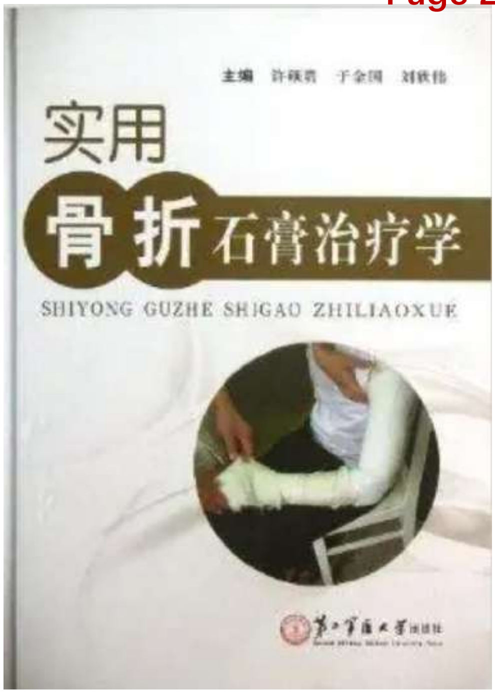
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骨折石膏治疗学
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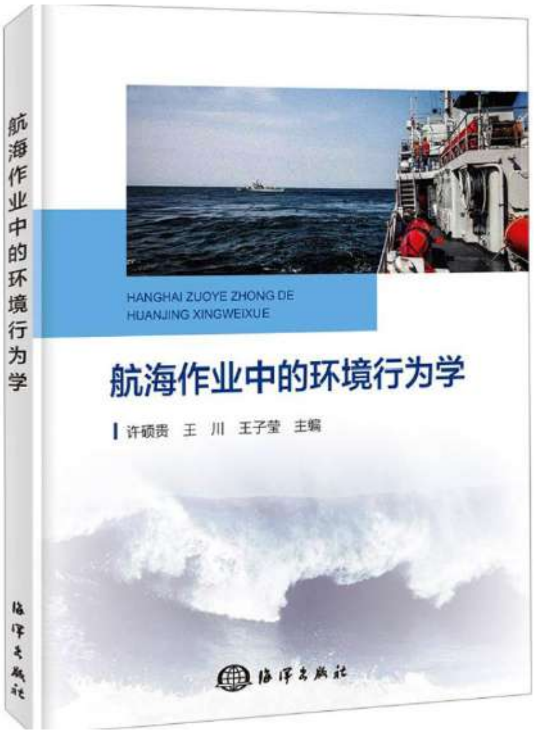
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